

Program : Diploma in Wood and Paper Technology	
Course Code : 4123	Course Title: Chemical Engineering
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide knowledge to the students with concepts of various unit operations and processes in wood based industries.
- To give an overview of mass transfer, heat transfer, fluid flow and mechanical operations.
- To apply material and energy balances and other basic laws in problem solving.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in mathematics		Mathematics I	1
Heat transfer, momentum, viscosity		Applied physics I	1
		Applied physics II	2
Concentration of solutions		Applied chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply the laws on mass and energy conservation in various industries.	14	Applying
CO2	Apply principles of mass transfer operations in process industries.	15	Applying

CO3	Discuss heat transfer operations and their applications in the design of equipments used in process industries.	16	Applying
CO4	Summarize the characteristics of particulate solids and various unit operations involving fluids and solid particles.	13	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3	2					
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Apply the laws on mass and energy conservation in various industries.		
M1.01	Differentiate unit operations and processes.	2	Understanding
M1.02	Recall the expression for composition of gas mixture and solutions and various governing laws.	6	Understanding
M1.03	Express Molal humidity, Absolute humidity, Relative humidity, percentage humidity, dew point, wet and dry bulb temperature	2	Understanding
M1.04	Solve material and energy balances problems	4	Applying

Contents:

Material Balance and Energy Balance

Unit operations and processes

Concept of mole, mole fraction, weight fraction, molality, molarity, normality

Ideal Gas law and Daltons law, partial pressure, Vapor Pressure

Ideal and non ideal solutions, Raoult's law, Henry's law Molal humidity, Absolute humidity, Relative humidity, percentage humidity, dew point, wet and dry bulb temperature Material balance- Law of conservation of mass – Steady and Unsteady state, Batch and continuous , material balance of equation for unit operations. Energy balance – Law of conservation of energy, Heat capacity, Sensible heat, Latent heat, Enthalpy, Internal Energy, Humid heat, Energy balance equation for unit operations			
CO2	Apply principles of mass transfer operations in process industries		
M2.01	Explain the mechanism of extraction , drying and absorption	5	Understanding
M2.02	Discuss the humidification process and use psychrometric chart	4	Applying
M2.03	Explain the principles of distillation and its types	4	Understanding
M2.04	Illustrate the working of tray column and packed tower for various operations.	2	Applying
	Series Test – I	1	
Contents Mass Transfer Extraction- liquid- liquid extraction, selection of solvent, solid - liquid extraction. Humidification- Humidification and dehumidification operation; adiabatic saturation temperature, psychrometric chart – industrial water cooling towers Drying- theory and mechanism, cylindrical drier Absorption - Selection of solvent, stripping Distillation – Relative volatility, Boiling point diagram ,Equilibrium diagram, simple distillation, equilibrium distillation, fractional distillation, Fractionation or continuous distillation with reflux, azeotropes, steam distillation Packed Tower and Tray(Plate) column			
CO3	Discuss heat transfer operations and their applications in the design of equipments used in process industries.		
M3.01	Explain the modes of heat transfer	4	Understanding
M3.02	State the basic laws governing heat transfer	4	Understanding
M3.03	Explain the phase change phenomena of boiling and condensation	4	Understanding

M3.04	Sketch the basic methodology of operation of heat transfer equipments	4	Applying
Contents: Heat Transfer Conduction- thermal conductivity, Fourier's law; Convection- natural & forced, heat transfer coefficient, Newtons law of cooling; Boiling and Condensation- mechanism and types Radiation- absorptivity, emissivity, reflectivity, Concept of black body, Laws of radiation- steffanboltzmanns law, kirchoffs law Heat Exchangers-flow pattern, LMTD, 1-1 shell & tube, double pipe Evaporators- single and multiple effect, short tube evaporator – falling film evaporator- forced circulation evaporators, cascade evaporators, Disc evaporators ;Steam economy calculation in evaporators			
CO4	Summarize the characteristics of particulate solids and various unit operations involving fluids and solid particles.		
M4.01	Discuss about different size reduction equipment using laws	4	Understanding
M4.02	Explain filtration, sedimentation, mixing	6	Understanding
M4.03	Interpret the fluid flow in industries and the ability to select valves by understanding its principles	3	Applying
	Series Test – II	1	
Contents: Fluid particle mechanics Size reduction- Laws of crushing and grinding, Crushing and grinding equipments Mixing and agitation Filtration- principle- filter types- construction and operation of rotary drum vacuum filter, sand filter Centrifugation – simple basket type centrifuge and Sedimentation. Flow through pipes – pipe fittings, gaskets and packing materials and valves Gate valve, globe valve, ball valve, butterfly valve, diaphragm valve, check valve, safety valve			

Text / Reference:

T/R	Book Title/Author
T1	McCabe W.L., Smith J.C. and Harriott P., “Unit Operations of Chemical Engineering”, 7th Ed., McGraw Hill.
R1	Narayanan K.V. and Lakshmikutty B., “Stoichiometry and Process Calculations”, Prentice Hall of India.
R2	Treybal R.E., “Mass Transfer Operations”, 3rd Ed., McGraw Hill..
R3	Badger and Banchero., “Introduction to Chemical Engineering”, IstEd., McGraw Hill, New York, 1954.
R4	Binay K. Dutta, “Heat Transfer Principles and applications” Prentice Hall of India Pvt. Ltd.
R5	S. K. Ghosal, S. K., Sanyal and S. Datta, “Introduction to Chemical Engineering”, Tata McGraw Hill Education Pvt. Ltd., New Delhi.

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112108149/
2	https://books.google.co.in/books/about/Unit_Operations_II.html?id=cyKSIfFJYfQC&redir_esc=y
3	https://books.google.co.in/books/about/Unit_Operations_i_Fluid_Flow_and_Mechani.html?id=K4almhE5BoAC&redir_esc=y